Figures and tables

Arnav Kapoor

IISER Bhopal.

Abstract

All figures and tables of "Exact, hardware-aware local rewriting of quantum circuits, with certified ground truth", regenerated from the results in this repository. Values shown as ?? come from runs that have not finished.

Equivalence as a group action

Table 1 Equivalence notions as group actions. Notions compose. For example, output-aware equivalence modulo chip symmetries uses row-normalised matrices minimised over the automorphism group.

notion	$U \sim V$ iff	group acting	canonical key
exact	$V = U$	trivial	U
up to phase [1]	$V = e^{i\varphi}U$	$U(1)$	phase-normalised U
input-aware	$V\|0\rangle = e^{i\varphi}U\|0\rangle$	stabiliser of $\|0\rangle$	phase-normalised $U\|0\rangle$
output-aware	$V = DU$, D diagonal	diagonal unitaries	rows phase-normalised
symmetry	$V = PUP^\dagger$	coupling-graph automorphisms	minimum over the orbit
approximate	$\|U - V\| \leq \varepsilon$	(a tolerance)	exact key, verified distance

Table 2 Hashed databases of the general optimiser (classes per equivalence notion).

device	topology	notion	depth	classes	build time
Heron	line	up to phase	9	6.31M	6 min
Heron	line	output-aware	10	0.34M	<1 min
Heron	line	input-aware	14	1.41M	<1 min
ion	triangle	up to phase	6	28.4M	38 min
ion	triangle	output-aware	6	7.36M	27 min
ion	triangle	input-aware	7	31.6M	23 min
paper ion ($\pm\frac{\pi}{2}$)	triangle	up to phase	9	2.93M	7 min
paper IBM ($\pm\frac{\pi}{2}$)	triangle	up to phase	10	1.84M	2 min

Certified ground truth

Table 3 Exact Cayley-graph data of the Clifford groups generated by the alphabets of [1].

alphabet	n	$\lvert\Gamma\rvert$	$\lvert G\rvert$	diameter	mean $\lvert g\rvert$ (uniform)
ion	2	14	11,520	10	5.89
IBM-style	2	9	11,520	11	6.92
ion	3	24	92,897,280	17	10.72
IBM-style	3	15	92,897,280	19	12.34

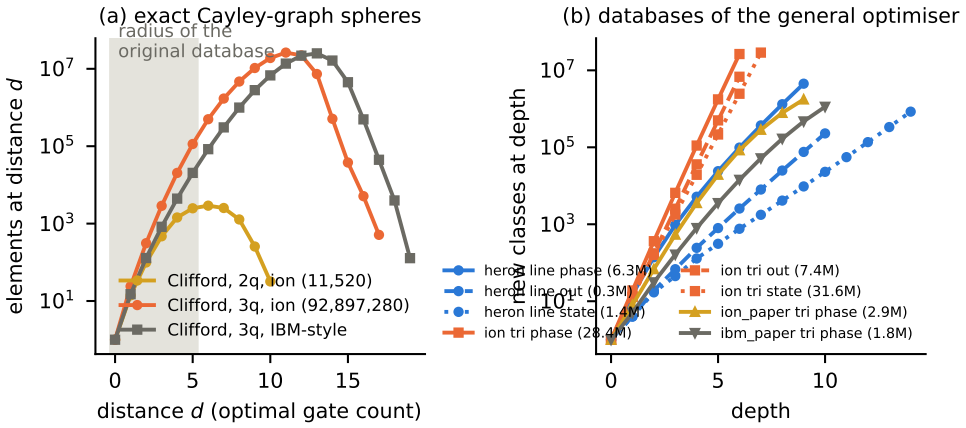


Fig. 1 (a) Sphere sizes of the exact Cayley graphs used as ground truth. The shaded band is the radius of the original database. The Clifford+T sphere sizes keep growing roughly tenfold per level, since the group is infinite. (b) Level sizes of the general optimiser's databases (keys per depth, one curve per device, topology and equivalence notion).

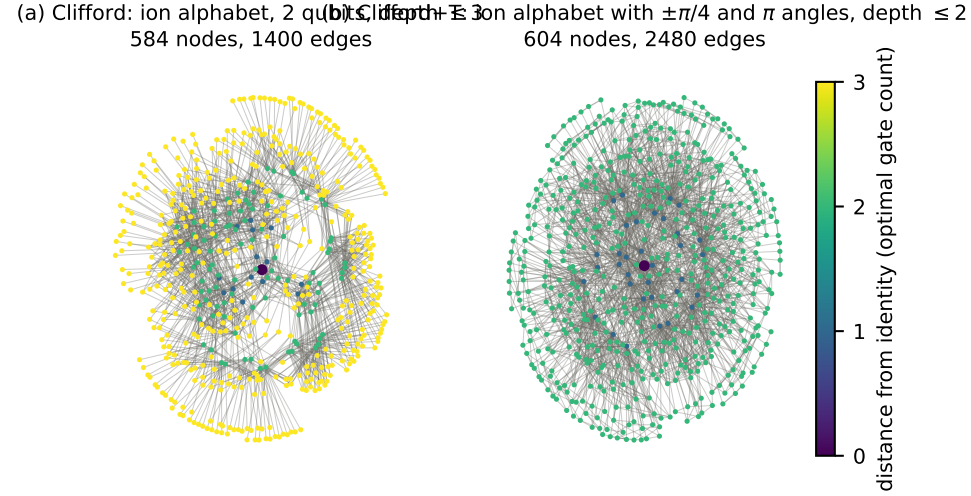


Fig. 2 Compute graphs of comparable size, nodes coloured by exact distance from the identity. (a) Two-qubit Clifford group, ion alphabet, depth ≤ 3. Many short words coincide, and the ball closes up. (b) Two-qubit Clifford+T, depth ≤ 2. Almost every word is new.

Experiments

Table 4 Mean gate counts on the original benchmarks. *Original and Qiskit at 4 qubits use the first 30 circuits. Qiskit is level 3, orig. is the original method, and Cliff. CW uses adaptive commutation-aware windows with the exact Clifford tables (Appendix B), which the general optimiser does not use. Optimum is the proven optimum or a certified lower bound, and #opt. counts the circuits on which the general optimiser is optimal. All 250/250 outputs verified and native.

setting	Qiskit	orig.	Cliff. CW	general	optimum	#opt.
T3, ion, 3q	127.2	47.1	–	10.8	10.4	17/20
T4, IBM-style, 3q	104.0	36.8	–	12.5	12.2	18/20
journal, ion, 4q	164.9*	78.8*	45.6	46.1	≥11.1	–
journal, IBM-style, 4q	140.9*	76.0*	49.3	51.4	≥12.9	–
T5, IBM-style, 15q	332.1	335.9	324.6	327.7	–	–

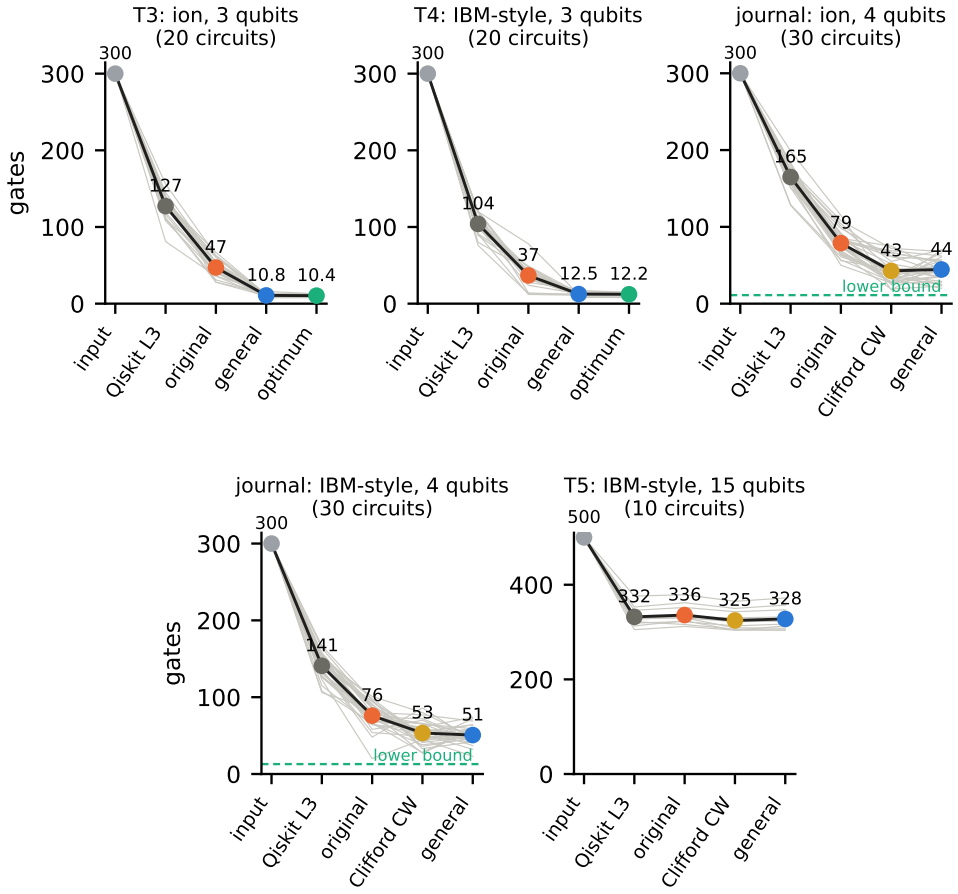
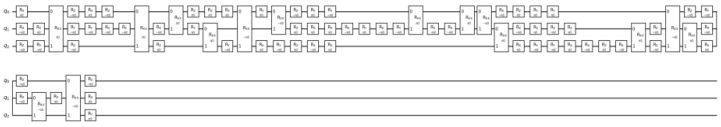


Fig. 3 From the input to the output of every method on the original benchmarks. Grey lines are individual circuits (paired by seed) and the black line is the mean. The dashed line at 4 qubits is the certified lower bound of Proposition 8.

(a) input: random 3-qubit word over the ion alphabet, 80 gates



(b) original method (database radius 5): 24 gates



(c) proven optimum (exact Cayley table): 11 gates



Fig. 4 A random 3-qubit word over the ion alphabet (a), the output of the original method (b) and the proven optimum from the exact Cayley table (c).

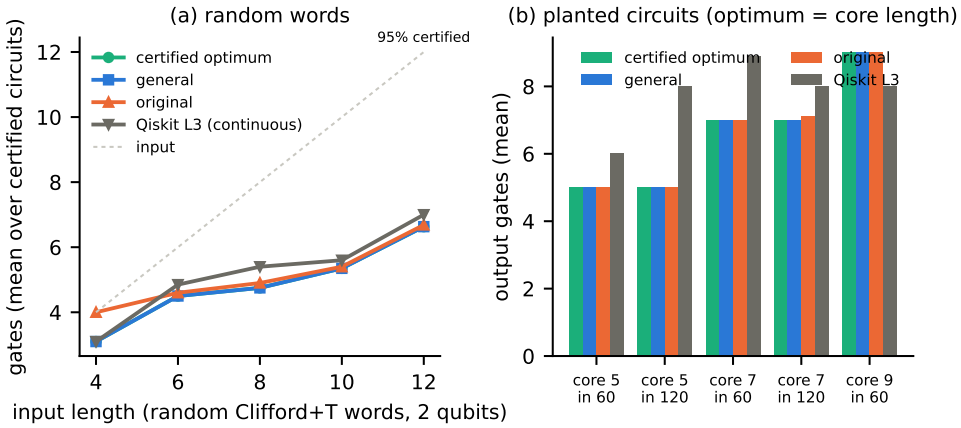
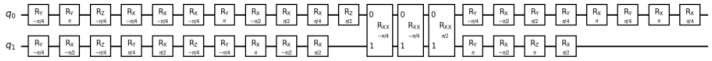


Fig. 5 Clifford+T, two qubits. (a) Random words, mean output length over circuits whose optimum is certified (annotations give the fraction certified). (b) Planted circuits, a core of the stated length hidden in 60 or 120 gates.

(a) planted Clifford+T circuit: 36 gates hiding a 5-gate optimum



(b) general optimiser: 5 gates (certified optimum: 5)

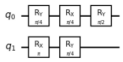


Fig. 6 A planted Clifford+T circuit (a) and the output of the general optimiser (b), which equals the certified optimum.

Table 5 Relative to Qiskit level 3 (58 circuits per device). Count columns are ratios of summed counts, and ESP is the geometric mean of per-circuit ratios. BQSKit outputs are approximate. Nam pf1_10/pf2_10 are excluded because Qiskit level 3 drops their $\sim 10^{-7}$ rad rotations (approximately, not exactly).

	IBM Heron			trapped ion		
pipeline	2q	all	ESP	2q	all	ESP
TKET (FullPeepholeOptimise on L1)	×1.02	×1.71	×1.16	×1.27	×1.79	×0.992
ours, exact, from L1	×1.00	×0.84	×1.21	×0.95	×0.76	×1.002
ours, exact, from L3	×0.98	×0.87	×1.03	×1.00	×1.00	×1.001
ours, approximate ($\varepsilon = 10^{-3}$), from L1	×1.01	×0.84	×1.21	×0.95	×0.75	×1.002
ours, state/measurement-aware, from L1	×0.77	×0.61	×1.28	×0.62	×0.50	×1.010
BQSKit level 2 on L1 (approximate)	×0.89	×1.26	×1.21	×0.96	×0.74	×1.002
BQSKit level 3 on L1 (approximate)	×0.85	×1.20	×1.22	×0.94	×0.71	×1.002

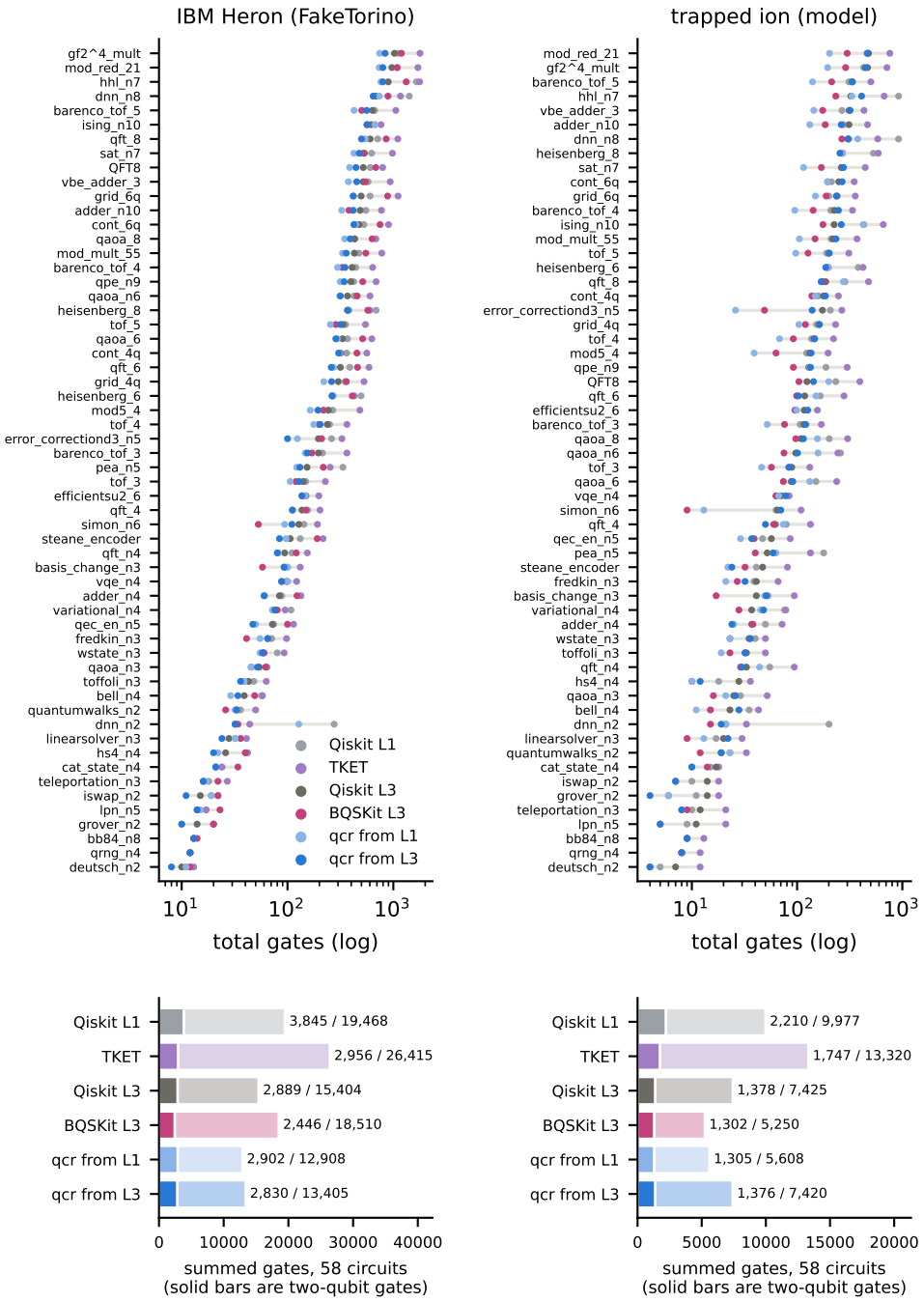
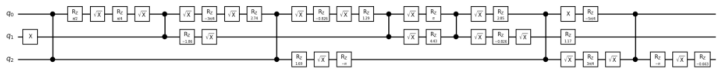


Fig. 7 Hardware suite, from the level-1 compilation to every pipeline's output. The top panel shows total gates per circuit (sorted by the level-3 count), the bottom panel the sums over all circuits, with two-qubit gates as solid bars.

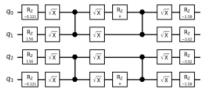
(a) toffoli_n3 after Qiskit level 3 on IBM Heron: 43 gates, 7 CZ



(b) toffoli_n3 after qcr (exact incl. global phase): 36 gates, 7 CZ



(c) hs4_n4 after Qiskit level 3 on IBM Heron: 26 gates, 4 CZ



(d) hs4_n4 after qcr (exact incl. global phase): 20 gates, 4 CZ

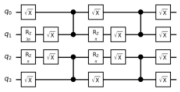


Fig. 8 Two QASMBench circuits after Qiskit level 3 on IBM Heron (a, c) and after `qcr` (b, d). The outputs are equal including global phase.

Table 6 Mean noisy-simulation TVD by objective (lower is better).

	Qiskit L1	Qiskit L3	ours, count	ours, $w2$	ours, calibrated
Heron	0.115	0.137	0.096	0.088	0.088
ion	0.0157	0.0091	0.0092	0.0084	0.0084

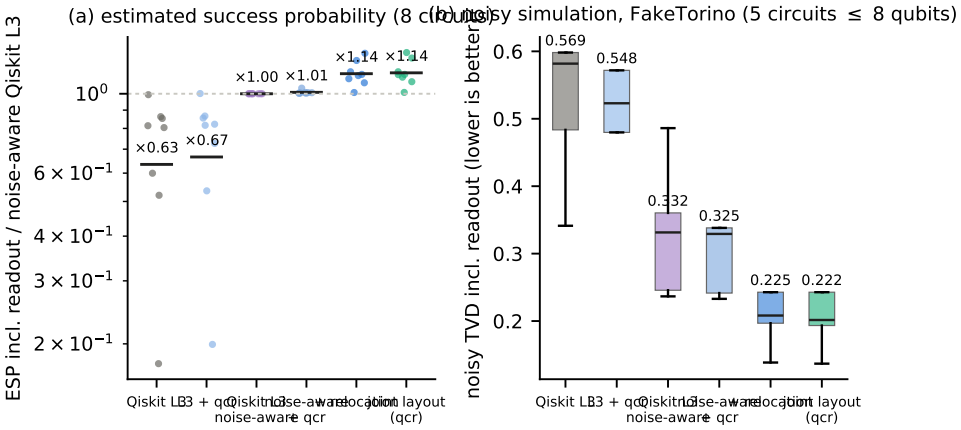


Fig. 9 Layout-aware rewriting on IBM Heron (FakeTorino). (a) ESP including readout relative to Qiskit's noise-aware level 3 (dots are circuits, bars and labels are geometric means). (b) Exact noisy TVD including readout for circuits with at most eight active qubits (labels: means).

Table 7 Layout-aware rewriting on IBM Heron. ESP includes readout, and ratios are geometric means relative to Qiskit's noise-aware level 3. TVD: mean exact noisy TVD including readout (5 circuits).

pipeline	ESP ratio	two-qubit gates (sum)	TVD
Qiskit L3 (coupling map only)	×0.63	493	0.569
Qiskit L3, noise-aware	×1.00	493	0.332
noise-aware + `qcr`	×1.01	485	0.325
noise-aware + `qcr` + relocation	×1.14	485	0.225
joint layout selection (`qcr`)	×1.14	471	0.222

Appendix: Reproduction of the reference method

Table 8 Reproduction of [1].

claim	status	detail
Table I (compute graphs)	exact	identical node and edge counts
Figs. 1, 4, 5 (examples)	reproduced	Fig. 4 requires a different gate pool (Theorem 6)
Tables III/IV, journal 4q	qualified	only with the 24-operator, depth-5 database, which is not stated
Table II (V1 > V2 > V3)	qualified	V3 wins only with a tuned forest threshold (recall 30% at the default)
Table V (15 qubits)	not reproduced	ties a current Qiskit (335.9 vs. 332.1 gates)
Fig. 9 (hardware)	qualified	present at Qiskit levels 0/1, absent at level 3

Appendix: Clifford-specific optimisers

Table 9 Mean gate counts in the setting of Tables III/IV of [1] (3 qubits, 300 gates, 20 circuits).

	optimum	original	EW-8	EW-16	Qiskit L3	paper (1 circuit)
ion	10.4	47.1	28.3	10.4	127.2	45
IBM-style	12.2	36.8	51.1	12.2	104.0	72
time per circuit (ion)	–	51 s	0.54 s	0.24 s	<0.1 s	–

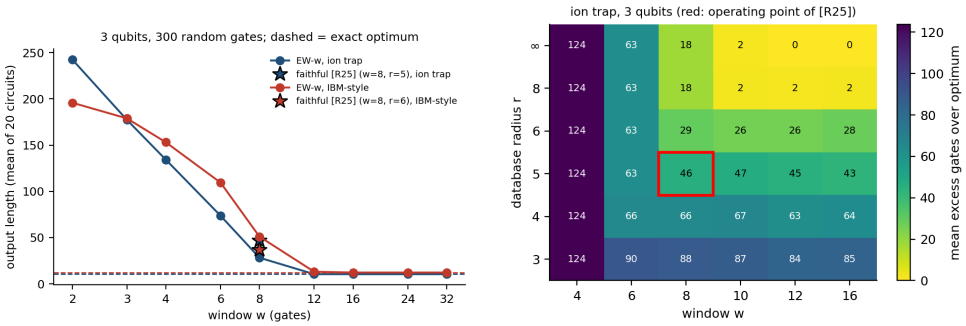


Fig. 10 Left, the fixed-point length of EW-w against w, with the cascade transition near $w^\star$. Right, the mean excess over the optimum by window w and database radius r (ion), with the original operating point $(8, 5)$ marked.

Table 10 Journal benchmark (4 qubits, 300 gates, 100 circuits). Original (orig.) and Qiskit level 3 use the first 30 circuits. Published (publ.) numbers are from the journal version of [1]. BQSKit level 4 is our run on the first ?? circuits, with the same gate set at continuous angles (approximate, all-to-all).

	bound	CW-sem	CW-24	EW-16	orig.	publ.	Qiskit	BQSKit
ion, total	$\geq$11.1	45.6	56.7	110.3	78.8	111	164.9	??
ion, RXX	–	–	27.6	44.2	33.3	43	47.3	??
IBM-style, total	$\geq$12.9	49.3	59.1	130.9	76.0	107	140.9	??
IBM-style, CZ	–	–	27.1	47.1	32.4	43	47.3	??

References

[1] Rosenhahn, B., Osborne, T.J., Hirche, C.: Optimization driven quantum circuit reduction. New Journal of Physics **27**, 104509 (2025). arXiv:2502.14715